fn match_group_by_truncation

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `match_group_by_truncation` in `mod.rs` at commit f5bb719 (outdated[1]).

1 Hoare Triple

Precondition

Compiler Verified

Types matching pseudocode.

Precondition

None

Function

```
def match_group_by_truncation(
    plan: DslPlan, identifier: Expr
) -> Optional[Tuple[DslPlan, Truncation, Bound]]:
    if not isinstance(plan, DslPlan.GroupBy): #
        return None

    input = plan.input
    keys = plan.keys
    aggs = plan.aggs
    apply = plan.apply
    options = plan.options

    if apply is not None or options != GroupbyOptions.default():
        return None #

    ids, by = partition(lambda expr: expr == identifier, keys) #

    if not ids: #
        return None

    return (
        input,
        Truncation.GroupBy(keys, aggs),
        Bound(by=by, per_group=1, num_groups=None), #
```

[1] See new changes with `git diff f5bb719..cc813c7 rust/src/transformations/make_stable_lazyframe/truncate/matching/mod.rs`

```
25     )
26
27
28 # part of Rust's standard lib, included for readability, with hardcoded types
29 def partition(
30     predicate, iterable: Iterable[Expr]
31 ) -> Tuple[HashSet[Expr], HashSet[Expr]]:
32     true_set = set()
33     false_set = set()
34     for item in iterable:
35         if predicate(item):
36             true_set.add(item)
37         else:
38             false_set.add(item)
39     return true_set, false_set
```

Postcondition

Theorem 1.1 (Postcondition)**.** For a given query plan and user identifier expression, if the query plan bounds row contributions per-identifier via a group-by, returns a triple containing the input to the truncation, the truncation itself, and the per-id bound on user contribution.

Proof. Lines 4-14 check if the query plan is a simple group-by, and exits without a match if not.

A valid group-by truncation is a group-by where the grouping keys contain a user identifier. So line 16 splits the group-by keys into two sets, and line 18 rejects the match if the user identifier is not in the group-by keys.

The remaining grouping columns are then reflected in the bound on user contributions, on line 24. □